

Feature Selection Vs Feature Extraction

Dimensionality Reduction

① Why Dimensionality Reduction?

3d 2d

* Prevent → Curse of Dimensionality

100d

* Improve the performance of the model

3d or 2d

* Visualize the data → understand the data

Feature Selection

I/P

x

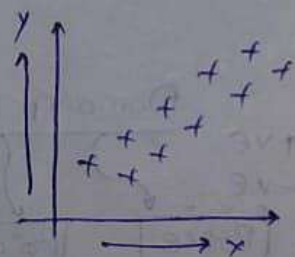
O/P

y

+ve ⇒

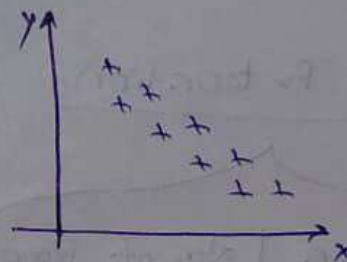
x↑ y↑
x↓ y↓

⇕



x↓ y↑
x↑ y↓

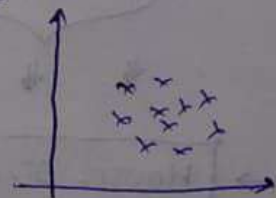
← -ve



$$\text{Cov}(x, y) = \sum_{i=1}^n \frac{(x_i - \bar{x}) * (y_i - \bar{y})}{N - 1} = +ve \approx 0 = -ve$$

No relation between x & y

$$\text{Pearson Correlation} = \frac{\text{Cov}(x, y)}{\sigma_x * \sigma_y} = [-1 \text{ to } 1]$$

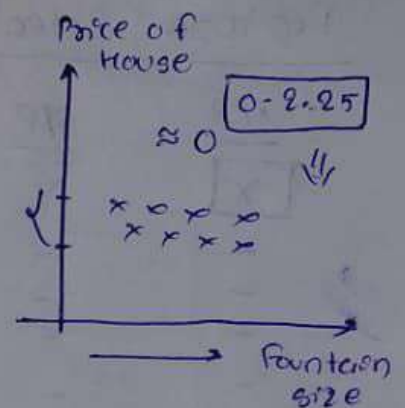
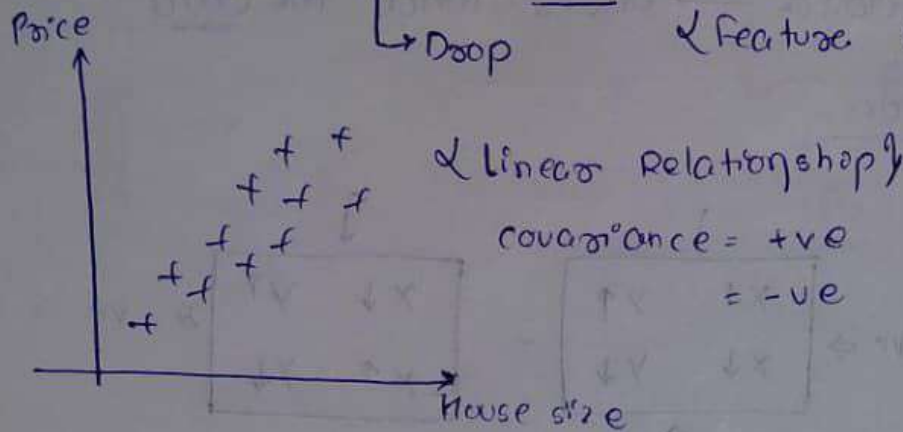
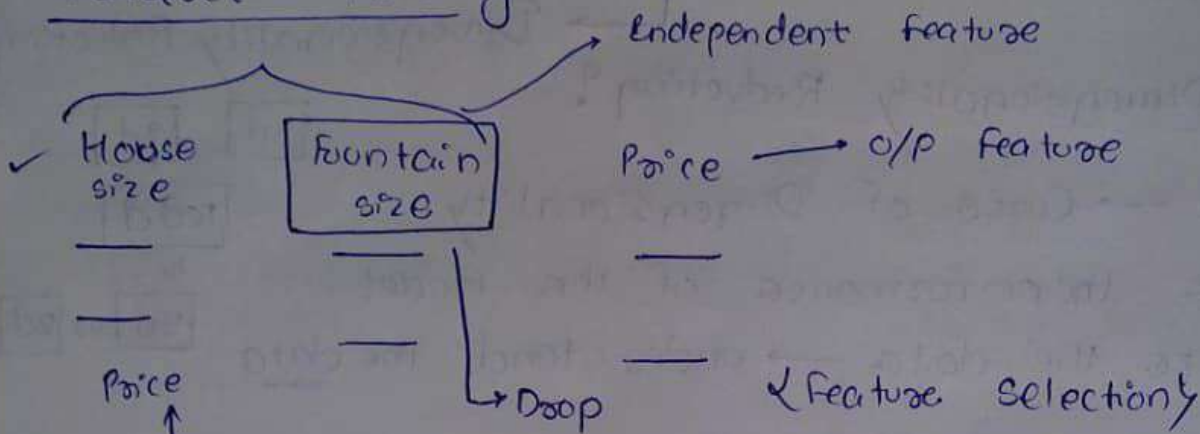


≈ 0 (No relation)

-ve correlated

The more towards the value of +1 the more +ve correlated x & y is.

Dataset Housing



Feature Extraction

